

Question number	Answer	Mark
1(a)(i)	The only correct answer is D. <i>A is incorrect because P is a ribosome (the smallest structure), Q is a starch grain (the largest structure)</i> <i>B is incorrect because P is a ribosome (the smallest structure), Q is a starch grain (the largest structure)</i> <i>C is incorrect because P is a ribosome (the smallest structure), Q is a starch grain (the largest structure)</i>	(1)

Question number	Answer	Mark
1(a)(ii)	The only correct answer is B. <i>A is incorrect because GALP is found in the stroma which is structure R</i> <i>C is incorrect because GALP is found in the stroma which is structure R</i> <i>D is incorrect because GALP is found in the stroma which is structure R</i>	(1)

Question number	Answer	Additional guidance	Mark
1(a)(iii)	14 267 / 14 270 / 14 300 / 14 400 / 14 000 (1)	ACCEPT in standard form e.g. 1.4267×10^4 DO NOT ACCEPT with units	(1)

Question number	Answer	Additional guidance	Mark
1(a)(iv)	An answer that includes at least one similarity and one difference: Similarities: - both have a phospholipid bilayer (1) Differences: - T has {chlorophyll / photosynthetic pigments / photosystems / PSI / PSII} in it but U does not (1) - T contains {ATP synthase / ATP ase} but U does not (1) - T contains electron carrier proteins but U does not (1)	DO NOT PIECE TOGETHER ACCEPT made of phospholipids ACCEPT electron carriers / ETC NB penalise once if refer to membranes or chloroplast as a cell	(3)

Question number	Answer	Additional guidance	Mark
1(b)	An answer that includes the following points: - light is needed for the replication (of chloroplast DNA) (1) - replication (of chloroplast DNA) is independent of {mitosis / cell cycle} (1)	ACCEPT replication of chloroplast DNA does not occur in the dark IGNORE DNA increases day / night	(2)